PROCEEDINGS B

royalsocietypublishing.org/journal/rspb

CC BY

Research

Cite this article: Karita Y, Rodríguez-Sánchez GT, Brambilla E, Hernandez-Beltran JCR, Schwarz M, Rainey PB. 2026 Context-dependent adaptation in structured environments. *Proc. R. Soc. B* **293**: 20252004.
https://doi.org/10.1098/rspb.2025.2004

Received: 4 August 2025
Accepted: 18 December 2025

Subject Category:
Evolution

Subject Areas:
evolution, microbiology

Keywords:
experimental evolution, structured environment, spatial dynamics, *Pseudomonas fluorescens*, air–liquid interface, bacterial mat, inoculum effect

Authors for correspondence:
Yuya Karita
e-mail: karita@noneq.phys.s.u-tokyo.ac.jp
Paul B. Rainey
e-mail: rainey@evolbio.mpg.de

Electronic supplementary material is available online at https://doi.org/10.6084/m9.figshare.c.8229719.

THE ROYAL SOCIETY PUBLISHING

Context-dependent adaptation in structured environments

Yuya Karita[1,2], Gisela T. Rodríguez-Sánchez[2], Elisa Brambilla[2], J. Carlos R. Hernandez-Beltran[2], Michael Schwarz[2] and Paul B. Rainey[2,3]

[1]Department of Physics, The University of Tokyo, Tokyo, Japan
[2]Department of Microbial Population Biology, Max Planck Institute for Evolutionary Biology, Plön, Germany
[3]Laboratory of Biophysics and Evolution, ESPCI, Université Paris Sciences et Lettres, Paris, France

YK, 0000-0001-8581-3630; GTR-S, 0009-0005-8998-5612; JCRH-B, 0000-0003-4791-7206; MS, 0009-0002-1414-3291; PBR, 0000-0003-0879-5795

Adaptive evolution often leads to niche specialization, but successful colonization of a new niche can depend as much on ecological context as on genetic change. This is especially true in spatially structured environments, where the construction of habitat and order of arrival of mutants shape evolutionary outcomes. In static broth cultures, the air–liquid interface (ALI) provides a high-oxygen niche typically colonized by mat-forming strategists. In *Pseudomonas fluorescens*, single mutations that upregulate cellulose production give rise to 'wrinkly spreader' (WS) types—canonical niche specialists that out-compete ancestral types at the ALI. Here, we show that this view can be refined: WS mutants often fail to establish when introduced alone at low density, but co-culture with ancestral types rescues colonization. Microscopy and simulations reveal that ancestral types physically scaffold the ALI, enabling attachment and facilitating the *in situ* emergence of diverse WS mutants. Ancestral types later disperse, allowing WS expansion. This transient asymmetric interaction shows that even strongly adaptive mutations may rely on ecological facilitation. More broadly, our findings highlight the reciprocal links between ecology and evolution, revealing how transient ecological dependencies can shape adaptive trajectories.

1. Introduction

Colonization of new habitats is a fundamental process shaping ecological and evolutionary dynamics across scales—from the establishment of plants on newly exposed land to microbial growth at fluid interfaces. Success depends on traits that allow organisms to arrive, attach and proliferate under local environmental constraints. Classical ecological theory identifies dispersal ability, rapid growth and efficient resource exploitation as key predictors of colonization success [1–4], while more recent frameworks recognize that cooperative traits and facilitative interactions can also determine which lineages persist and dominate [5–9]. Despite extensive theory, the mechanisms by which these principles manifest at the microbial scale remain only partly understood.

The air–liquid interface (ALI) in static cultures provides a tractable model for studying colonization in a spatially structured, resource-stratified environment. This oxygen-rich niche supports the adaptive evolution of mat-forming bacteria [10–14]. Aerobic growth quickly depletes oxygen in the liquid bulk, creating steep vertical gradients that favour mutants capable of reaching and exploiting the meniscus [15]. Once established, these lineages produce cohesive mats that secure access to oxygen and exclude competitors—an outcome broadly analogous to niche construction or ecosystem engineering in larger-scale systems.

© 2026 The Authors. Published by the Royal Society under the terms of the Creative Commons Attribution License http://creativecommons.org/licenses/by/4.0/, which permits unrestricted use, provided the original author and source are credited.

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

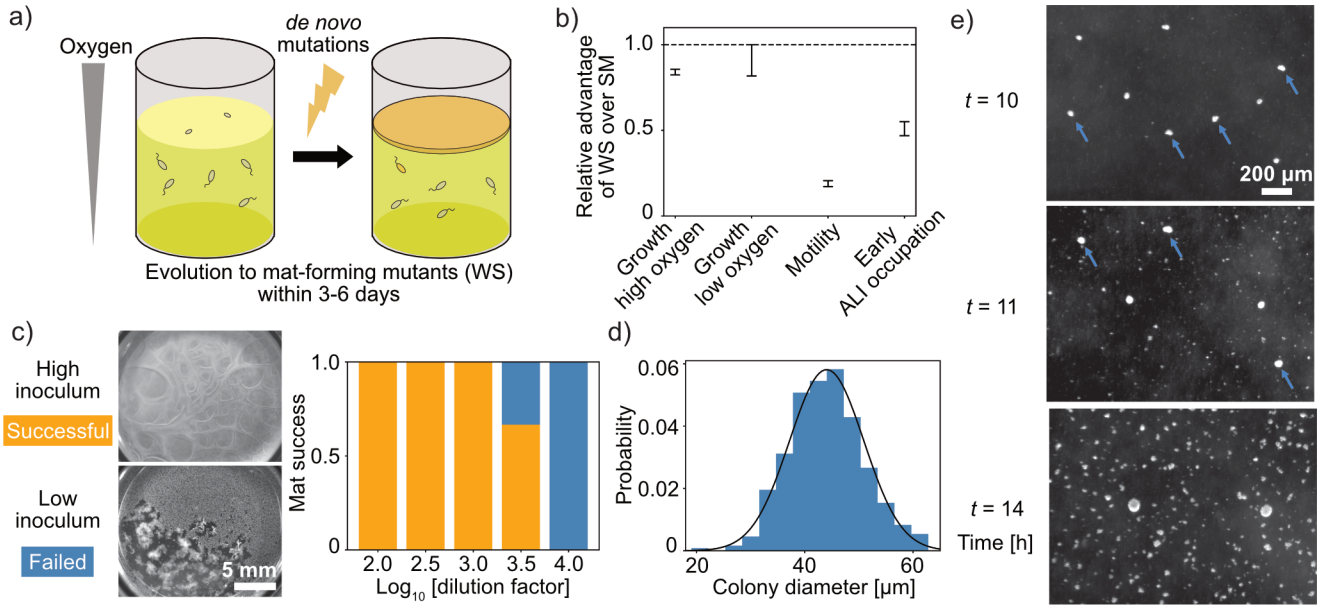


Figure 1. Adaptive mutants exhibit context-dependent fitness limitations. (a) *P. fluorescens* SBW25 reproducibly evolves mat-forming WS mutants in static culture within 3–6 days. (b) Summary of WS phenotypes compared to the ancestral SM type. The ratio of a quantified trait of WS over that of SM was plotted for each advantageous trait. Growth rate data are from Koza *et al.* [15]. Motility was measured using a semi-solid agar assay. Early ALI occupation was quantified as the ratio of surface cell density at $t = 2$ h to the initial inoculum density. (c) WS mutants failed to form a mat in a monoculture when initiated from low inoculum densities. Mat formation success was scored at $t = 24$ h across a dilution series. Orange bars show the ratio of successful populations, and blue bars depict the ratio of failed populations. $N = 3, 6, 12, 15, 9$ for each dilution factor (from 10^2 to 10^4). (d) Distribution of WS microcolony sizes immediately prior to collapse in monoculture ($N = 769$ colonies). Diameter was calculated from the measured colony area, assuming a completely circular shape. The black curve indicates a Gaussian fit. (e) Dark-field images showing microcolony collapse events in WS monocultures. Blue arrows mark detached colonies falling from the ALI.

Pseudomonas fluorescens SBW25 has been a model for understanding this process. Under static conditions, the ancestral smooth morphotype (SM) cannot stably occupy the ALI, but de novo mutants known as 'wrinkly spreaders' (WS) evolve reproducibly within days [10](figure 1a). WS types overproduce extracellular matrices due to mutations that elevate cyclic dimeric guanosine monophosphate (c-di-GMP), a second messenger controlling the motile–sessile transition in many bacterial species [16–18]. These mutations occur via multiple genetic pathways at a combined rate of approximately 1.1×10^{-8} per generation [19], ensuring rapid and parallel evolutionary responses. The over-secreted extracellular matrices work as structural glues for connecting cellular clusters and confer the robustness and coherence of ALI mats.

WS mutants exhibit several fitness trade-offs that complicate their success (figure 1b). They grow more slowly than the ancestral SM under both oxygen-rich and oxygen-poor conditions [15,20], and their reduced motility limits migration towards the ALI [21,22]. However, WS types are more hydrophobic than SM [23], allowing them to raft stably once they reach the surface. The resulting fitness landscape is defined by a trade-off between impaired migration to, and enhanced persistence at, the ALI [21]. Complicating matters further, ancestral SM types can transiently colonize the interface by forming ephemeral microcolonies that coalesce and collapse within hours [24]. Together, these observations challenge the view that WS mutants simply out-compete ancestors through superior colonization ability.

Previous work has proposed that WS lineages succeed by generating collective benefits—specifically, access to oxygen through mat formation—despite individual costs [7,20]. This process can be interpreted as a microbial analogue of ecological succession: the ancestral SM initially modifies the environment in ways that facilitate the establishment of WS, which in turn replaces its progenitor. Unlike classical cases of succession in which later colonizers immigrate from elsewhere, here the successor arises *de novo* from the resident population through mutation. The transition therefore fuses ecological and evolutionary timescales, producing a successional sequence generated internally rather than by species turnover.

In this study, we revisit WS evolution from an ecological perspective. We show that WS types, when introduced alone at low density, often fail to colonize the ALI within 24 h. This deficiency is rescued by ancestral SM cells, revealing a facilitative interaction between genotypes. Microscopy and modelling demonstrate that SM cells physically scaffold the interface, compensating for WS motility defects and enabling stable attachment. Once established, WS types subsequently displace SM through differential dispersal dynamics—a 'dispersal-driven' transition that contrasts with classical selective sweeps. These findings expose a hidden dependency on ancestral types in what appears to be a simple case of adaptive radiation, and more broadly link microbial evolution at interfaces to general ecological principles of facilitation and succession operating across levels of organization.

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

2. Results

(a) Fitness costs associated with WS types

To quantify fitness costs during early ALI colonization, we measured the surface cell density at $t = 2$ h after inoculation and calculated the ratio of surface cell density and inoculum density to quantify the efficiency of ALI colonization for monocultures of various genotypes (figure 1b, 'Early ALI occupation'; electronic supplementary material, figure S1). A representative WS mutant (caused by deletion of *wspF*; see electronic supplementary material, table S3) showed lower surface densities than SM, indicating reduced colonization efficiency. Similar trends were observed for other WS genotypes (*awsX* and *mwsR* mutants; electronic supplementary material, table S3), although some showed slightly higher efficiency than SM (electronic supplementary material, figure S1). None, however, displayed a clear advantage that could explain the rapid and reproducible evolution of WS.

A more striking result emerged when we further varied the inoculum density. WS mutants failed to form a mat when cultured alone from low inoculum densities ($\sim 10^6$ cells ml^{-1}, figure 1c; electronic supplementary material, video S1). Consistent with previously reported delayed mat formation of *Bacillus subtilis* at low density [25], mat failure occurred during the initial 24 h colonization window and was not observed in subsequent rounds of mat formation, which proceeded successfully regardless of inoculum size. This finding raises an important question: how do spontaneous WS mutants in laboratory evolution colonize the ALI so reliably if they cannot establish from low density?

To explore this paradox, we used microscopy to track ALI microcolony formation at low inoculum densities. In WS monocultures, microcolonies repeatedly collapsed once they reached a diameter of approximately 40 µm, producing hole-like defects that ultimately prevented coherent mat formation (figure 1d,e; electronic supplementary material, video S3).

These collapses are consistent with a size-dependent physical instability of floating microcolonies. The mass densities of bacterial cells (approx. 1.1 g cm^{-3} [26,27]) and extracellular matrices (in the case of cellulose, it is approximately 1.5 g cm^{-3} or approximately 1.37 g cm^{-3} in an amorphous state [28,29]) exceed that of the surrounding medium. For small colonies below the capillary length (on the order of millimetres), surface tension provides the primary support. However, surface tension scales with length, while gravitational force scales with volume. As a result, microcolonies become unstable once they exceed a critical size, consistent with our observations.

To formalize this dynamic, we developed a spatial model simulating early ALI colonization (see electronic supplementary material for details). The model distinguishes between a well-mixed bulk phase and a structured ALI, represented as a two-dimensional lattice. Colonization proceeds via local growth and aggregation of surface-attached microcolonies. Successful mat formation requires that colonies connect to form a coherent mat before reaching the size threshold for collapse (figure 2c,e). The resulting dynamics are characterized by sharp shifts in success rate as a function of inoculum density (figure 2c,d), resembling a percolation transition [30], in which the number of connected clusters increases drastically. Simulations also revealed that traits enhancing surface dispersal—such as motility—significantly increase colonization success (figure 2b,d), underscoring the disadvantage of WS mutants in this early phase.

(b) Compensation for WS fitness costs

The failure of WS mutants to form mats at low inoculum raises a fundamental question: what enables their repeated success in evolution experiments? A key difference between these monoculture assays and standard evolution protocols is the presence of the ancestral SM type. We hypothesized that SM cells might play a direct role in facilitating WS establishment.

To test this, we performed co-culture experiments. When SM cells were added to low-inoculum WS populations (conditions under which WS alone failed to form a mat), mat formation was restored in approximately 20% of replicates (2 out of 9). Microscopy revealed that SM and WS cells co-colonized the ALI (figure 3a). Based on insights from our spatial model (figure 2), we infer that the presence of SM stabilizes early ALI colonization by helping to connect neighbouring microcolonies, thereby preventing collapse. These results suggest a positive mechanical interaction between the two types, which has not been previously reported.

To investigate the nature of this interaction, we tracked microcolony growth in co-cultures using fluorescence microscopy. When mixed at equal frequency, WS and SM jointly colonized the ALI, and no clear differences in growth or expansion were observed (figure 3b). This indicates that the positive effect of SM on WS establishment is not due to a direct fitness gain for WS, but rather emerges from the physical properties of collective colonization.

We next examined the role of ancestral SM cells in the evolutionary dynamics via laboratory evolution of a monoculture of genetically engineered SM. We used an SM genotype carrying a c-di-GMP-responsive reporter (pCdrA–mScarlet) [31,32]. The reporter is non-fluorescent under basal conditions but becomes fluorescent when intracellular c-di-GMP levels rise, allowing tracking of the emergence and expansion of WS mutants, which over-produce c-di-GMP.

When this genotype was cultured under static conditions, fluorescence was undetectable during early growth but became visible after the ALI was fully colonized by non-fluorescent SM cells. At $t = 24$ h, we detected a small fluorescent colony at the ALI, and by $t = 48$ h, multiple fluorescent microcolonies had appeared (figure 4a). These foci resembled locally growing WS clusters, consistent with their reduced motility and limited dispersal. To confirm that fluorescence reflected WS identity, we plated evolved ALI populations on agar plates. Multiple colony morphologies typical of WS were recovered, most showing fluorescence. In contrast, fluorescent SM colonies were rarely observed (figure 4b,c). The existence of dark WS and fluorescent SM colonies is probably explained by sensitivity of the reporter [32,33].

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

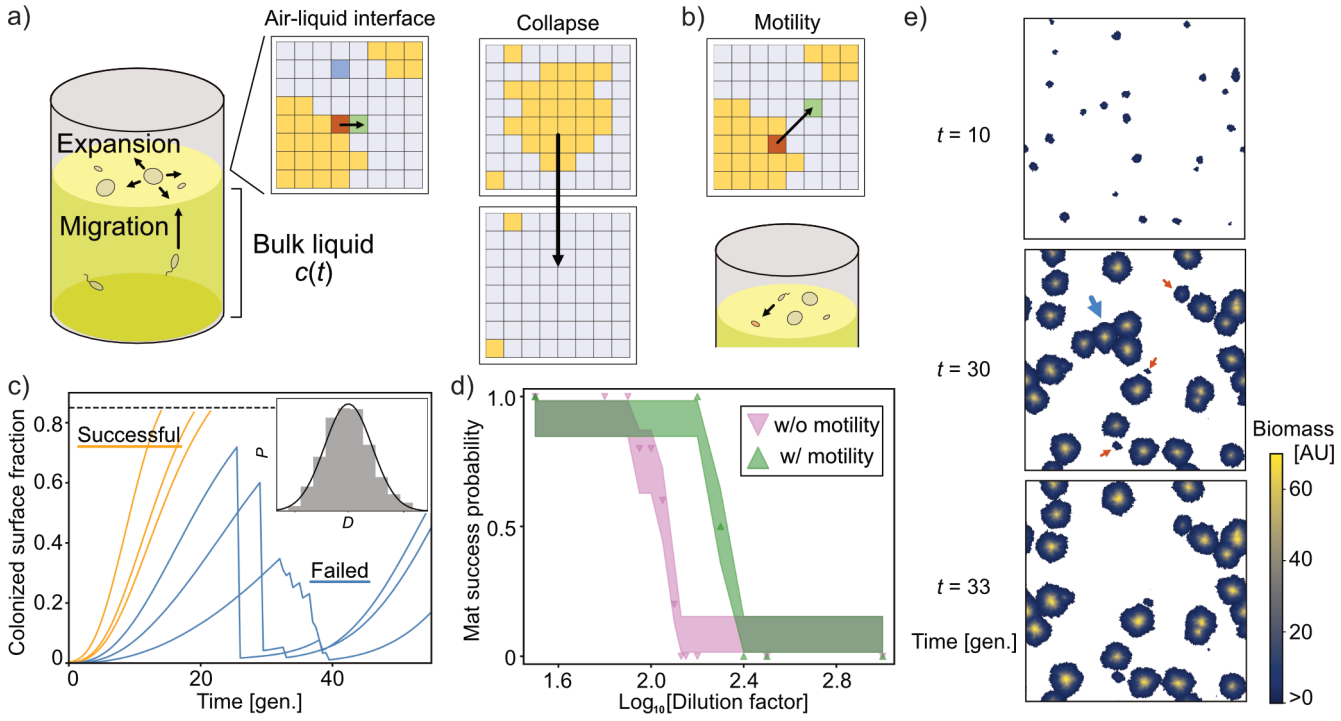


Figure 2. Spatial simulations reveal inoculum-dependent success in ALI colonization. (a,b) Schematics of the simulation model. The culture is divided into a well-mixed bulk liquid and a structured ALI represented as a 2D lattice. Microcolonies grow, merge and collapse. Collapsing occurs based on size-dependent physical instability. See electronic supplementary material for model details. (c) Example simulation trajectories without motility. Mat formation was successful at higher inoculum densities (orange lines; dilution factors $10^{1.5}$–$10^{1.9}$), but failed at lower inocula (blue lines; dilution factors $10^{2.2}$–10^{3}). When 85% of the ALI was covered (dotted line), simulations were terminated, and mat formation was regarded as successful. The inset shows the size distribution of collapsing isolated colonies (*D*: diameter in an arbitrary unit. *P*: normalized frequency); the solid line is a Gaussian fit. (d) Probability of successful mat formation by $t = 30$ [generation] as a function of inoculum density, with (green) and without (pink) motility. Shaded regions show 68% CIs from a beta distribution ($N = 10$ replicates per condition). (e) Simulation snapshots at a low inoculum (dilution factor 10^{3}). Early colonizers attach at $t = 10$; later colonization occurs via migration from the bulk (red arrows at $t = 30$). Large aggregates collapse due to instability (blue arrow at $t = 30$).

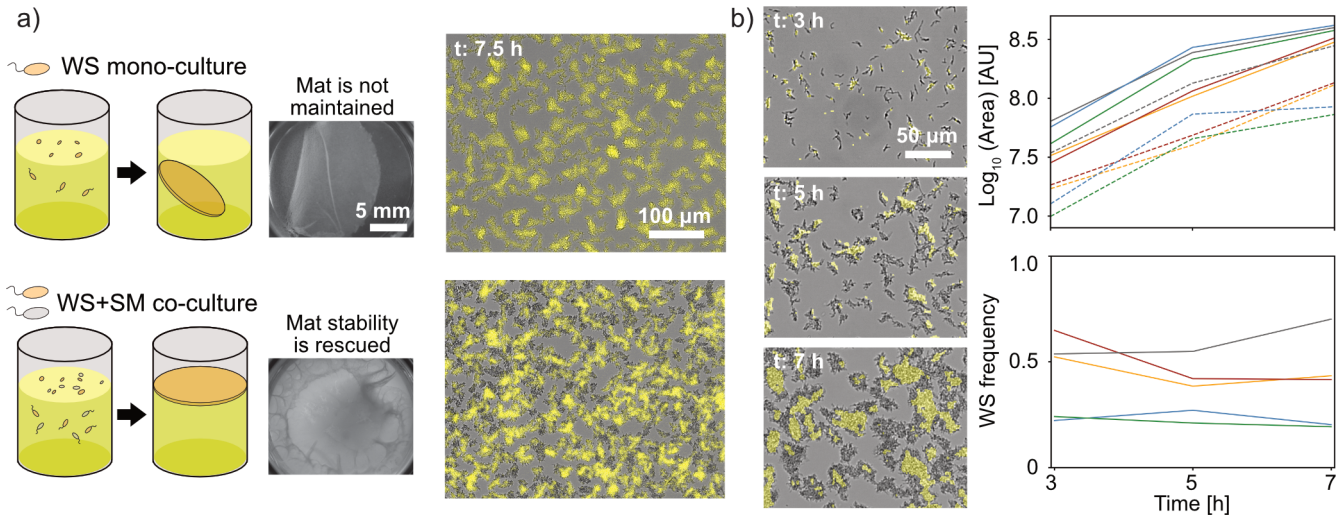


Figure 3. Ancestral SM cells enable WS colonization through mechanical facilitation. (a) Mat formation failed in WS monocultures at low inoculum density (top row), but was rescued in approximately 20% of replicates (2 out of 9) when a similar number of SM cells were co-inoculated (bottom row). Compared with WS monoculture (0 out of 15 were successful), the difference in the success rate of mat formation was significant up to 76% CI of a beta distribution (see §4). Middle panels show representative top-views of a failed (top) and stable (bottom) mat. Right panels show ALI colonization at $t = 7.5$ h, with WS visualized using a fluorescent marker (shown in yellow). (b) Fluorescence time-lapse microscopy of WS/SM co-cultures revealed no clear differences in expansion rate or competitive fitness at the ALI. The top plot shows the area expansion of the total population (solid lines) and WS population (dashed lines). The change in WS frequency is shown in the bottom plot. Line colours denote experimental replicates. The fluorescent tag was flipped in blue and green replicas.

Together, the c-di-GMP reporter experiments reveal the spatio-temporal sequence of WS evolution: (i) SM transiently colonizes the ALI, (ii) WS mutants emerge locally within the SM population at the ALI, and (iii) WS expands via local growth. Pre-colonization by SM appears to act as a physical scaffold that enables newly emerging WS cells to remain surface-associated and avoid microcolony collapse. Moreover, the presence of SM at the ALI increases the likelihood of WS mutations arising directly at the interface. This spatial positioning offsets the motility disadvantage of WS mutants, as they no longer need to migrate from

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

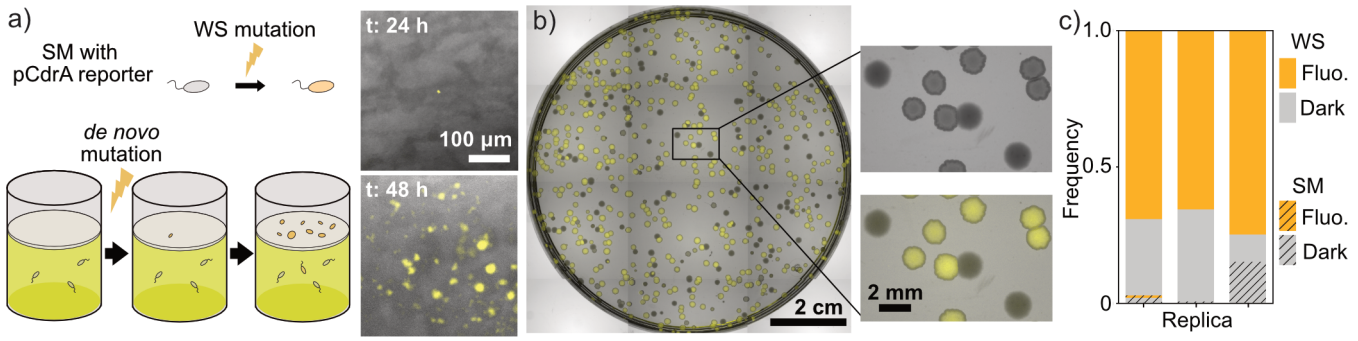


Figure 4. Spatial dynamics of WS emergence revealed by a c-di-GMP reporter. (a) Ancestral SM cells carrying a pCdrA–mScarlet reporter were evolved in static culture. Fluorescence was undetectable early on but appeared around $t = 24$ h, after the ALI was fully colonized by non-fluorescent cells ($t \sim 10$ h). By $t = 48$ h, multiple fluorescent microcolonies were visible across the ALI. (b) After 6 days of evolution, cells from the ALI were plated on agar plates. WS and SM types were distinguished by colony morphology. Fluorescence was primarily observed in WS colonies. (c) Frequency of colony morphotypes and fluorescence expression across replicate lines. More than 1000 colonies were scored per treatment.

the bulk to access the niche. These results highlight the central role of ancestral populations in facilitating adaptive evolution of WS phenotypes.

(c) How slow-growing WS out-competes SM

While our results show that SM cells can compensate for the fitness costs of WS mutants during colonization, this alone does not explain how WS types ultimately increase in frequency and dominate the ALI population. Given that WS mutants grow more slowly than SM, what allows their eventual dominance?

We hypothesized that the answer lies in the general transition between sessility and dispersal in biofilm-forming bacteria [17]. Many bacterial species regulate motility and extracellular matrix production in response to environmental conditions, often using c-di-GMP as a central signalling molecule. During biofilm development, cells frequently shift to a motile state to disperse once the biofilm matures [34]. In *P. fluorescens*, WS mutations elevate c-di-GMP levels [22], locking cells into a sessile state and reducing their capacity to disperse.

If SM cells are more prone to dispersal, then WS may gain an indirect advantage simply by remaining localized at the ALI (figure 5a). To test this, we tracked motility transitions in a SM-majority population. Initially, SM cells formed a weak mat that covered the ALI by $t = 10$ h. Shortly after, many cells became motile, leading to phase separation between active fluid regions and sessile microcolonies (see electronic supplementary material, video S4). Using particle image velocimetry (PIV), we observed dynamic fluid motion within these motile regions over a 60 min interval (figure 5b). In contrast, WS clusters remained largely sessile and stable.

To formalize this dynamic, we developed a four-state model incorporating two cell types (SM and WS) and two spatial locations (ALI and bulk liquid) (figure 5c). The population size of each state changed over time via growth, spatial replacement and mutation of cells. WS was assigned lower growth and migration rates than SM, reflecting its fitness disadvantages. Its sole advantage was a reduced tendency to disperse from the ALI. Assuming differential oxygen availability between the ALI and the bulk, and a carrying capacity constraint at the interface (see electronic supplementary material for details), the model reproduced WS takeover, despite disadvantages to individual cells (figure 5d,e).

The model simulations are consistent with our c-di-GMP reporter experiment (figure 4), showing the *in situ* emergence and expansion of WS cells at the ALI upon SM pre-colonization. Notably, the model also predicts a steep increase in the number of newly arising WS mutations at the ALI following a rapid increase of the ALI population size (figure 5e,f). This suggests the emergence and coexistence of multiple adaptive WS lineages in a single population.

(d) Diversity of WS lineages in a single evolved population

The evolutionary dynamics of WS mutants in static cultures depart markedly from the classical selective sweep model, in which a single high-fitness lineage fixes in the population. WS mutants exhibit multiple fitness disadvantages and depend on the presence of ancestral SM cells to establish at the ALI. Their spread is driven less by intrinsic growth rate than by reduced dispersal, suggesting that colonization success may not require competitive exclusion.

These features raise the possibility that multiple WS lineages could emerge and coexist within a single evolving population. To test this, we performed evolution experiments using a 1 : 1 mixture of two SM genotypes, each carrying a different fluorescent tag. After 6 days of static culture, the resulting mat populations were plated on agar plates, and individual colonies were scored for both colour and morphology (figure 6a).

WS colony morphologies were classified into three types—small, large and wheel-like—following established criteria [35]. Across five replicate evolution lines, all six combinations of colour and morph (2 colours × morphs) were consistently observed (figure 6b,c), confirming the presence of multiple distinct WS lineages within each population. Since different genotypes can produce similar morphologies [19,36], the true genotypic diversity of WS mutants is probably even higher.

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

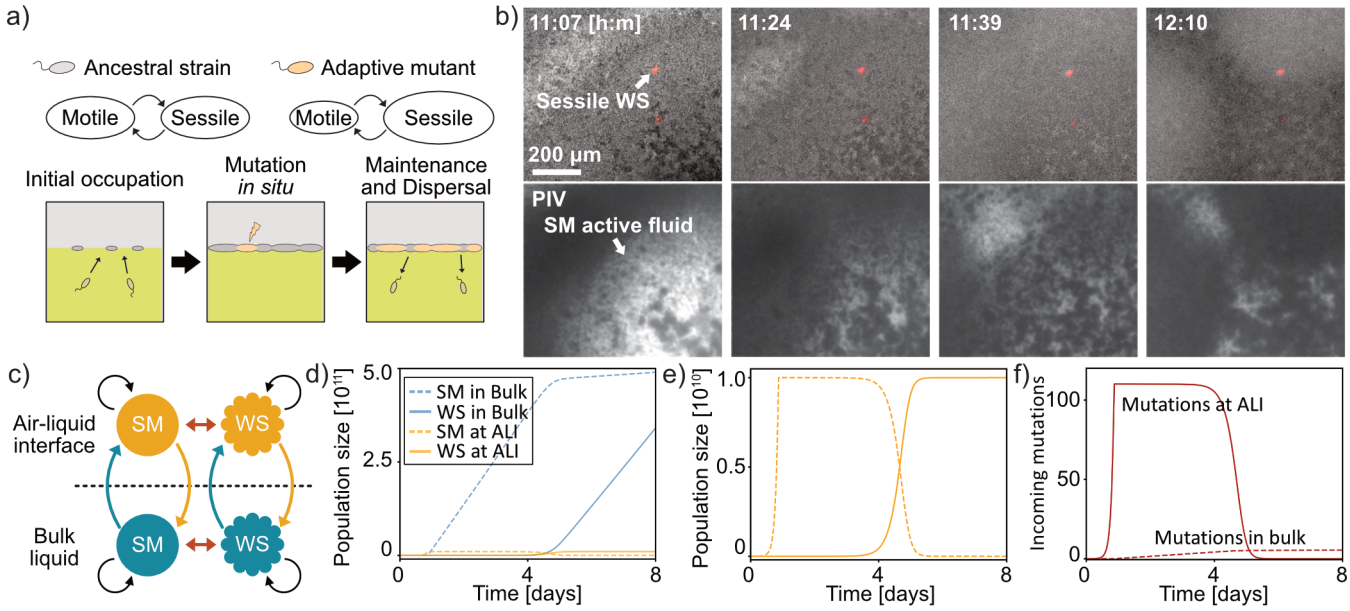


Figure 5. Dispersal-driven dynamics enable WS takeover despite growth disadvantages. (a) A conceptual schematic of the dispersal-driven takeover. SM cells exhibit greater motility and are more likely to disperse from the ALI, while WS cells remain sessile and persist at the interface. (b) Transition to motility in an SM-majority population. By $t = 11$ h, active fluid regions emerged as SM cells became motile. Top panels show the stability of rare WS clusters (red), while bottom panels show the velocity of SM active fluid at the same location. The velocity field was calculated by particle image velocimetry (PIV), and the magnitudes ($|v|$) were time-averaged over 5 second intervals to visualize the local activity. (c) Schematic of a four-state model with two genotypes (SM and WS) and two spatial environments (ALI and bulk liquid). The four states develop via growth (black arrows), migration from bulk to ALI (blue arrow), dispersal from ALI to bulk (orange arrows) and mutational transition (red arrows). WS has lower growth and migration rates than SM but benefits from reduced dispersal from the ALI. (d) Simulated population dynamics of each state over time. Despite lower intrinsic fitness, WS comes to dominate due to higher retention at the ALI. (e) Magnified population dynamics at the ALI. (f) The number of new mutations (from SM to WS) at the ALI increases steeply over time, reflecting the local emergence and coexistence of multiple adaptive WS lineages.

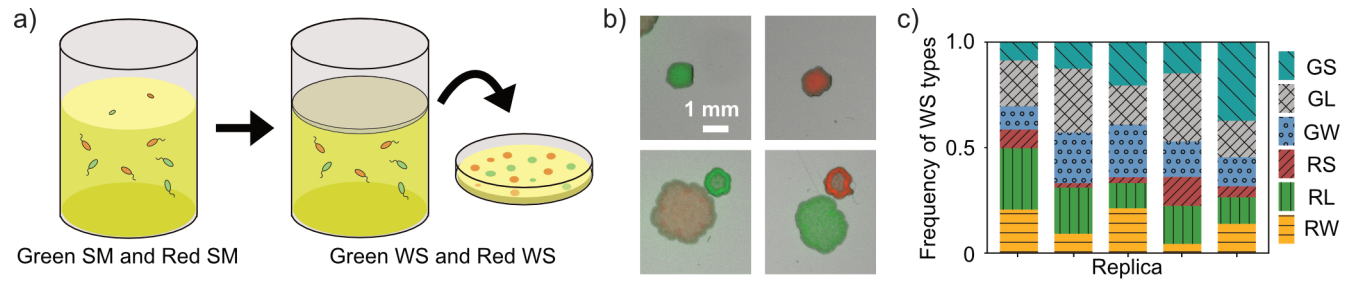


Figure 6. Coexistence of diverse WS lineages within evolved populations. (a) Schematic of the experimental set-up. Two ancestral SM genotypes, each tagged with a distinct fluorescent marker (green and red), were mixed 1 : 1 and evolved under static conditions for 6 days. The resulting mat populations were plated on agar plates to assess colony colour and morphology. (b) Representative image showing the six observed WS colony types, defined by combinations of colour and morphology. (c) Distribution of WS morphotypes across five independent evolution lines. More than 140 colonies were scored per replicate. All six morphotypes were consistently recovered, indicating the co-emergence and coexistence of multiple WS lineages within single evolved populations. Letters show colours (G: green; R: red) and morphs (S: small WS; L: large WS; W: wheel-like WS).

While a full analysis of WS diversity and its scaling with population size is beyond the scope of this study, such patterns could be explored using approaches such as deep sequencing or genetic barcoding, which have recently been developed for *P. fluorescens* SBW25 [37]. Our results nonetheless demonstrate that WS diversification is a consistent outcome of evolution in this system, shaped by spatial structure, ecological context and dispersal dynamics.

3. Discussion

The consistent emergence of WS mutants in static cultures of *P. fluorescens* SBW25 [10] has long served as a model for studying adaptive radiation and colonization of the ALI [38]. Despite their reproducible appearance, WS mutants possess inherent disadvantages, such as reduced growth rates [15,20] and impaired motility [21,22], raising questions about their evolutionary success. Previous work has attributed success to strain-specific properties of the mutants, such as overproduction of cellulose [39] and surfactant secretion [21]. Here, our study highlights the importance of ecological context [40–42], revealing that prior colonization of the ALI by ancestral smooth (SM) cells creates a physical scaffold that facilitates the establishment and eventual dominance of WS mutants.

This 'dispersal-driven' dynamic underscores a broader ecological principle: transitions between motility and sessility in response to environmental structure [17,43]. In typical biofilm life cycles, cells switch between motile and sessile states according

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

to local cues [34]. WS mutations, characterized by elevated c-di-GMP levels, lock cells into a sessile state that, while disadvantageous in fluctuating or well-mixed environments, becomes beneficial in the stable niche of the ALI. This explains the consistent emergence of WS mutants in laboratory settings.

Our findings also highlight a novel eco-evolutionary interaction mediated by a mechanical effect. The pre-colonization of the ALI by SM cells provides a physical scaffold that enables the establishment of WS mutants. This interaction is reminiscent of the evolution of auxotrophs, where ancestral cells produce metabolites essential for mutant survival [44,45]. However, in our system, the interaction is purely mechanical, arising from a trade-off between extracellular matrix production and motility. Such mechanical, or structural, facilitation may be widespread in mat-forming bacteria across diverse species and environments.

However, it is worth noting that mechanical facilitation may have negative consequences for populations. The excess of SM cells is known to weaken the structural coherence of mats [46]. In the more general context of biofilms, negative impacts of cells that do not produce extracellular products have been reported [47,48]. Thus, there exists a duality of ecological roles for the ancestral type: on the one hand, it facilitates establishment of mats, but at the same time, albeit at later stages, SM cells compromise structural integrity of WS mats.

Two features of our model system also warrant consideration. First, the oxygen gradient between the bulk liquid and the ALI is fundamental. SM occupation of the ALI prevents oxygen supply to the bulk, suppressing the growth and viability of mutants arising there. Consequently, *in situ* mutations at the ALI are more likely to succeed. This effect further supports the scenario that WS mutants arise at the ALI and rarely migrate to this niche from the bulk phase. Second, the relatively large carrying capacity at the ALI increases the probability of beneficial mutations occurring *in situ*. Given the size of our microcosms, the population size of an early-stage mat is estimated as 10^8–10^9 cells based on spatial coverage. For a mature mat, the carrying capacity is approximately 10^{10} cells (determined by breaking mats by sonication—see §4). These population sizes are comparable with population size in the bulk liquid, which is estimated as 10^8–10^9 cells given the reduced growth rate in the bulk due to oxygen depletion [15]. Considering that the typical combined mutation rate to WS phenotype is about 1.1×10^{-8} per generation [19], the ALI population size is large enough for evolutionary dynamics to occur in an experimentally reasonable time scale. However, we note that, in smaller niches, such as colonic crypts in mammalian guts, colonization by mutants may rely more on rare migration events from external populations rather than *in situ* mutations.

The question of whether migration or mutation occurs first in niche colonization is pertinent. Given the trade-off between motility and adhesion, mutations that enhance niche colonization often impair dispersal. Our study demonstrates that pre-colonization by ancestral cells can facilitate the establishment of such mutants. However, in scenarios where ancestors cannot pre-colonize the niche, two evolutionary pathways are conceivable: (i) a one-step mutation enables niche colonization followed by rare migration, or (ii) an initial mutation enhances motility, allowing niche occupation, followed by a second mutation that promotes sessility. The latter pathway may be more likely when migration is particularly challenging. Laboratory experiments support this sequence, showing the evolution of niche colonizers from ancestors lacking key components for ALI colonization, such as strains deficient in c-di-GMP pathways (Δ*wsp*Δ*aws*Δ*mws*) [36,49] or cellulose production (Δ*wss*) [50]. These evolved types often exhibit reduced motility, with mutations affecting chemotaxis or flagellar function [22,49]. Such observations support the plausibility of a two-step evolutionary process in niche colonization.

There exist comparable dispersal–colonization trade-offs across diverse taxa [43,51]. For example, in cavity-nesting birds, behavioural shifts between exploratory and territorial phases depend on population density and resource availability, mediating transitions between dispersive and resident strategies [52]. Similarly, in insects, the evolution of winged and wingless morphs reflects context-dependent selection on dispersal versus reproduction [53]. These examples, like the bacterial system studied here, illustrate how organisms balance the costs of movement with the benefits of local persistence, and how environmental structure and social or mechanical facilitation can determine which strategy prevails.

In summary, our findings reveal that the reproducible emergence of WS mutants is not solely a consequence of their intrinsic properties but is significantly influenced by the ecological context established by ancestral SM cells. The interplay between motility, adhesion and environmental structure exemplifies a general principle: successful colonization often depends on transient facilitative phases that bridge dispersal and settlement. Viewed in this light, the bacterial transition from SM to WS parallels broader patterns of ecological succession and dispersal–colonization trade-offs observed throughout the living world.

4. Methods

(a) Strains and culture conditions

P. fluorescens SBW25 strains (see electronic supplementary material, table S3 for the strain list) were cultured in King's B medium (KB) [54] at 28°C. To avoid the strong autofluorescence of KB in the green wavelength, mScarlet was used as the main fluorescent marker for bacteria. A *wspF* mutant [55] was used as a typical WS type. Prior to experiments, cells were streaked on an agar plate from a frozen stock and incubated for 2 days. To obtain a saturated cell suspension, cells from a single colony on a plate were taken and cultured in a liquid medium with continuous shaking for 24 h. When diluting or transferring cells, cell suspensions were agitated well with a vortex mixer for more than 1 min to resolve clumps.

(b) Measurements of motility and early ALI occupation

Swimming motility was quantified on freshly prepared semi-solid agar plates (0.25% agar and 1% KB, SSA) following the previously established method [32]. 200 µl of 24 h old shaken liquid cultures was transferred to wells in a 96-well plate after brief

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

vortexing. The cell suspensions were set down for approximately 1 h to let debris settle, and cells were carefully sampled with a toothpick from the top surface of the liquid. The toothpick was vertically stabbed in the centre of an SSA plate. SSA plates were incubated at 28°C for 24 h. The swimming halo diameter was measured with a digital vernier (Caliper).

To quantify early ALI colonization, cells at the ALI were observed by an upright microscope (Axio Imager Z2, Zeiss) with a 10× objective closely placed above the ALI. Starting cultures were inoculated in a 12-well microtitre plate (TPP) and incubated in a microscope incubator (PeCon). To measure the surface density, the number of cells at $t = 2$ h was directly counted around the centre of the ALI, where the effect of the air-liquid meniscus on the z focus is minimized. The measurement was iterated across various volume inoculum densities. The volume densities were quantified by plating diluted suspensions on agar plates and counting colonies. The ratio of the surface density to the inoculum density was defined as a measure of the efficiency of early ALI occupation.

(c) Observation of mat formation and surface colonization

A saturated liquid pre-culture was diluted to the required inoculum densities in a 12-well microtitre plate (TPP) to start static cultures. The macroscopic dynamics of mat formation were tracked by dark-field imaging with a stereo microscope (Axio Zoom V16, Zeiss) every 10 min for 24 h. During static cultures, a stage-top incubator with a heated lid (ibidi) was used to reduce evaporation. The timing of the mat collapse was manually detected from timelapse images. To analyse the critical size for a single colony to collapse, a magnified dark-field image (140×) was taken at the centre of an ALI, and the colony size at the time immediately prior to collapse was measured. The colony size was quantified as the area of connected pixels after applying the Otsu binarization and closing morphological operation. To observe the microscopic configuration of microcolonies at the ALI, an upright fluorescence microscope (Axio Imager Z2) was employed. To obtain time-resolved images, XY drift due to strong surface fluctuations of the ALI was compensated by manual XY adjustments.

(d) Evaluation of mat success rate

To evaluate the experimental statistics of mat success and failure, we assumed the dynamic to be a Poisson process with the success rate p. Given that a and b are the number of successful observations and failed observations, respectively, the likelihood of p should follow a beta distribution $B(a,b)p^a(1-p)^b$, where $B(a,b)$ was defined as $B(a,b) = \Gamma(a+1)\Gamma(b+1)/\Gamma(a+b+2)$ with a Gamma function Γ. The possible range of p was calculated from a beta distribution given a confidence interval D.

(e) Co-culture experiments for mat rescue

To test ecological interactions between WS and SM strains, we conducted co-culture experiments and checked the success rate of mat formation. We first confirmed that the intended inoculum density (approx. 2×10^6 cells ml^{-1} corresponding to approx. $10^{3.65}$ dilution) of WS did not lead to any successful colonization using 12 lines in a 12-well plate (TPP). Next, 3 wells of a 12-well plate were used to reconfirm no mat formation of WS monocultures, and 9 lines were used for cocultures of WS and SM (approximately 2×10^6 cells ml^{-1} for each). The subsequent ALI colonization dynamics were recorded with a stereo microscope (Axio Zoom V16) every 10 min for 24 h. For the statistical evaluation, a Poisson process with the success rate of mat formation p was assumed. The possible range of p was calculated using a beta distribution with confidence intervals D for WS monocultures and WS-SM cocultures, respectively. The maximum D such that the inferred range of p did not overlap for two cases was calculated.

(f) Evolution experiments

To track the spatio-temporal course of evolutionary dynamics, the ancestral SM genotype carrying a c-di-GMP-responsive reporter (pCrA-mScarlet) was cultured from a high inoculum density (approx. 10^7 cells ml^{-1}) for 6 days. 50 µg ml^{-1} Gentamicin was added to the culture media to maintain the plasmid and prevent contamination. The spatial expression of fluorescence was tracked with an upright microscope (Axio Imager Z2). To examine the fluorescence and the morphotype of evolved cells, the mat structure was carefully sampled from the ALI with a pipette. The suspension was centrifuged at a low speed (150 rcf) for 2 min (MiniSpin, Eppendorf). The supernatant was gently removed, and the sediment was resuspended in 1 ml PBS. The sample was sonicated for 15 s (Sonopuls mini20, Bandelin) to resolve the mat structure and plated at 10^6 and 10^7 dilutions. The colonies on plates were manually scored with a stereo microscope (Axio Zoom V16).

To evaluate the diversity of WS lineages in derived populations, a 1 : 1 mixture of the ancestral SM genotype carrying different fluorescent tags (sGFP2 and mScarlet) was statically cultured from a high inoculum (approx. 10^7 cells ml^{-1}) at 28°C. At 6 days, the evolved mat population was sampled and plated as described above. The fluorescence and the morphology of colonies were manually scored by a stereo microscope (Axio Zoom V16). The colony morphology of WS mutants was classified into three subclasses following the established criteria [35].

(g) Observation of motility transition at the ALI

To track the transition to motility at the ALI, a small number (approx. 10^3 cells ml^{-1}) of WS cells carrying a fluorescent tag (mScarlet) were mixed into an SM-majority (approx. 10^6 cells ml^{-1}) population. The suspension was cultured under a microscope incubator,

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

and the spatio-temporal dynamics were tracked with an upright microscope (Axio Imager Z2). At each time point, after manually correcting XY drift, a snapshot of fluorescence expression was captured, and the active movements were recorded for 5 s with bright-field imaging. To access the local activity, the XY drift was computationally corrected based on the feature pattern of an arbitrarily chosen sessile region, and PIV was conducted with PIVlab [56] in MATLAB (MathWorks). The XY drift was further corrected by subtracting the spatially averaged velocity of a sessile region. The magnitude of the velocity at each position was averaged over 5 s to quantify local activity.

(h) Model simulations

Early-phase colonization of the ALI was simulated with a spatial model that distinguishes between a well-mixed bulk phase and a structured ALI phase, represented as a two-dimensional lattice (see electronic supplementary material for details). Cells in the bulk liquid grow logistically and migrate to the ALI to colonize a random vacant position at a rate proportional to the bulk population size. At the ALI, colonies grow horizontally following the Eden model rule [57] and accumulate biomass locally. The stability of an ALI colony is evaluated by the net biomass and the colony periphery length, and unstable colonies are removed from the ALI. The stochastic simulations were conducted by the next reaction method [58,59] and terminated once 85% (chosen arbitrarily) of the ALI was covered by cells.

To formalize 'dispersal-driven' evolutionary dynamics, we developed a four-state model (see electronic supplementary material for details) with two cell types (SM and WS) and two spatial locations (bulk liquid and ALI). The time development of the four states is driven by growth, spatial relocation and genetic mutations. We assumed a carrying capacity of an ALI reflecting a spatial constraint (the finite ALI area). With this constraint, the dispersal rate from the ALI and migration rate to the ALI are assumed to be dependent on the ALI population size. To reflect genotype fitness, WS was assigned lower growth and migration rates than SM. The sole positive trait of WS is reduced dispersal rates from the ALI. Oxygen depletion is implemented as a reduction in the growth of both cell types in the bulk liquid. Simulations were founded by SM-only populations.

Ethics. This work did not require ethical approval from a human subject or animal welfare committee.

Data accessibility. The supplementary movies, spreadsheet data and scripts are available at Zenodo [60]. The microscopy data are available at BioImage Archive [61].

Supplementary material is available online [62].

Declaration of AI use. We have not used AI-assisted technologies in creating this article.

Authors' contributions. Y.K.: conceptualization, data curation, formal analysis, investigation, methodology, software, validation, visualization, writing—original draft; G.T.R.-S.: conceptualization, investigation, methodology, validation, writing—review and editing; E.B.: conceptualization, methodology, validation, writing—review and editing; J.C.R.H.-B.: conceptualization, software, validation, writing—review and editing; M.S.: investigation, methodology, validation, writing—review and editing; P.B.R.: conceptualization, project administration, supervision, validation, writing—original draft.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

Funding. This work was enabled by generous core support from the Max Planck Society.

Acknowledgements. We thank Andrew D. Farr, David Rogers, Maxime Ardré and Loukas Theodosiou for helpful discussion and comments, and Carsten Fortmann-Grote for advice on employing computational clusters and organizing the data repository. We acknowledge Ellen McConnell, Joanna Summers, Lena Meister and Michael Barnett for constructing genotypes in the Microbial Population Biology strain repository at the Department of Microbial Population Biology, Max Planck Institute for Evolutionary Biology, Germany. This work was enabled by generous core support from the Max Planck Society.

References

1. MacArthur RH, Wilson EO. 1967 *The theory of island biogeography*. Princeton, NJ: Princeton University Press.
2. Connell JH, Slatyer RO. 1977 Mechanisms of succession in natural communities and their role in community stability and organization. *Am. Nat.* **111**, 1119–1144. (doi:10.1086/283241)
3. Tilman D. 1994 Competition and biodiversity in spatially structured habitats. *Ecology* **75**, 2–16. (doi:10.2307/1939377)
4. Fukami T. 2015 Historical contingency in community assembly: integrating niches, species pools, and priority effects. *Annu. Rev. Ecol. Evol. Syst.* **46**, 1–23. (doi:10.1146/annurev-ecolsys-110411-160340)
5. Govaert L, Altermatt F, De Meester L, Leibold MA, McPeek MA, Pantel JH, Urban MC. 2021 Integrating fundamental processes to understand eco-evolutionary community dynamics and patterns. *Funct. Ecol.* **35**, 2138–2155. (doi:10.1111/1365-2435.13880)
6. Snell-Rood EC, Kobiela, ME, Sikkink, KL, Shephard AM. 2018 Mechanisms of plastic rescue in novel environments. *Annu. Rev. Ecol. Evol. Syst.* **49**, 331–354. (doi:10.1146/annurev-ecolsys-110617-062622)
7. Hammerschmidt K, Rose CJ, Kerr B, Rainey PB. 2014 Life cycles, fitness decoupling and the evolution of multicellularity. *Nature* **515**, 75–79. (doi:10.1038/nature13884)
8. Rose CJ, Hammerschmidt K, Pichugin Y, Rainey PB. 2020 Meta-population structure and the evolutionary transition to multicellularity. *Ecol. Lett.* **23**, 1380–1390. (doi:10.1111/ele.13570)
9. Doulcier G, Remigi P, Rexin D, Rainey PB. 2025 Evolutionary dynamics of nascent multicellular lineages. *Proc. R. Soc. B* **292**, 20241195. (doi:10.1098/rspb.2024.1195)
10. Rainey PB, Travisano M. 1998 Adaptive radiation in a heterogeneous environment. *Nature* **394**, 69–72. (doi:10.1038/27900)
11. Kovács ÁT, Dragoš A. 2019 Evolved biofilm: review on the experimental evolution studies of Bacillus subtilis pellicles. *J. Mol. Biol.* **431**, 4749–4759. (doi:10.1016/j.jmb.2019.02.005)
12. Pentz JT, Lind PA. 2021 Forecasting of phenotypic and genetic outcomes of experimental evolution in *Pseudomonas protegens*. *PLoS Genet.* **17**, e1009722. (doi:10.1371/journal.pgen.1009722)
13. Smith TM, Youngblom MA, Kernien JF, Mohamed MA, Fry SS, Bohr LL, Mortimer TD, O'Neill MB, Pepperell CS. 2022 Rapid adaptation of a complex trait during experimental evolution of *Mycobacterium tuberculosis*. *eLife* **11**, e78454. (doi:10.7554/eLife.78454)

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

14. Pentz JT, Biswas A, Alsaed B, Lind PA. 2024 Forecasting of phenotypic and genetic outcomes of experimental evolution in Pseudomonas syringae and Pseudomonas savastanoi. *Evol. Biol* 2024–02. (doi:10.1101/2024.02.10.579745)
15. Koza A, Moshynets O, Otten W, Spiers AJ. 2011 Environmental modification and niche construction: developing O2 gradients drive the evolution of the wrinkly spreader. *ISME J.* **5**, 665–673. (doi:10.1038/ismej.2010.156)
16. Hengge R. 2009 Principles of c-di-GMP signalling in bacteria. *Nat. Rev. Microbiol.* **7**, 263–273. (doi:10.1038/nrmicro2109)
17. Guttenplan SB, Kearns DB. 2013 Regulation of flagellar motility during biofilm formation. *FEMS Microbiol. Rev.* **37**, 849–871. (doi:10.1111/1574-6976.12018)
18. Valentini M, Filloux A. 2016 Biofilms and cyclic di-gMP (c-di-GMP) signaling: lessons from *Pseudomonas aeruginosa* and other bacteria. *J. Biol. Chem.* **291**, 12547–12555. (doi:10.1074/jbc.R115.711507)
19. Lind PA, Libby E, Herzog J, Rainey PB. 2019 Predicting mutational routes to new adaptive phenotypes. *eLife* **8**, e38822. (doi:10.7554/eLife.38822)
20. Rainey PB, Rainey K. 2003 Evolution of cooperation and conflict in experimental bacterial populations. *Nature* **425**, 72–74. (doi:10.1038/nature01906)
21. Jerdan R, Kuśmierska A, Petric M, Spiers AJ. 2019 Penetrating the air-liquid interface is the key to colonization and wrinkly spreader fitness. *Microbiology* **165**, 1061–1074. (doi:10.1099/mic.0.000844)
22. Giddens SR, Jackson RW, Moon CD, Jacobs MA, Zhang XX, Gehrig SM, Rainey PB. 2007 Mutational activation of niche-specific genes provides insight into regulatory networks and bacterial function in a complex environment. *Proc. Natl Acad. Sci. USA* **104**, 18247–18252. (doi:10.1073/pnas.0706739104)
23. Spiers AJ, Rainey PB. 2005 The *Pseudomonas fluorescens* SBW25 wrinkly spreader biofilm requires attachment factor, cellulose fibre and LPS interactions to maintain strength and integrity. *Microbiology* **151**, 2829–2839. (doi:10.1099/mic.0.27984-0)
24. Ardré M, Dufour D, Rainey PB. 2019 Causes and biophysical consequences of cellulose production by *Pseudomonas fluorescens* SBW25 at the air–liquid interface. *J. Bacteriol.* **201**, 10–1128. (doi:10.1128/JB.00110-19)
25. Hölscher T, Bartels B, Lin YC, Gallegos-Monterrosa R, Price-Whelan A, Kolter R, Dietrich LEP, Kovács ÁT. 2015 Motility, chemotaxis and aerotaxis contribute to competitiveness during bacterial pellicle biofilm development. *J. Mol. Biol.* **427**, 3695–3708. (doi:10.1016/j.jmb.2015.06.014)
26. Loferer-Krössbacher M, Klima J, Psenner R. 1998 Determination of bacterial cell dry mass by transmission electron microscopy and densitometric image analysis. *Appl. Environ. Microbiol.* **64**, 688–694. (doi:10.1128/AEM.64.2.688-694.1998)
27. Milo R, Phillips R. 2015 *Cell biology by the numbers*. New York, NY: Garland Science.
28. Chen W, Lickfield GC, Yang CQ. 2004 Molecular modeling of cellulose in amorphous state. Part I: model building and plastic deformation study. *Polymer***45**, 1063–1071. (doi:10.1016/j.polymer.2003.11.020)
29. Mazeau K, Heux L. 2003 Molecular dynamics simulations of bulk native crystalline and amorphous structures of cellulose. *J. Phys. Chem. B* **107**, 2394–2403. (doi:10.1021/jp0219395)
30. Davis S, Trapman P, Leirs H, Begon M, Heesterbeek JAP. 2008 The abundance threshold for plague as a critical percolation phenomenon. *Nature* **454**, 634–637. (doi:10.1038/nature07053)
31. Rybtke MT, Borlee BR, Murakami K, Irie Y, Hentzer M, Nielsen TE, Givskov M, Parsek MR, Tolker-Nielsen T. 2012 Fluorescence-based reporter for gauging cyclic Di-GMP levels in Pseudomonas aeruginosa. *Appl. Environ. Microbiol.* **78**, 5060–5069. (doi:10.1128/AEM.00414-12)
32. Summers J. 2023 Evolution of developmental regulation in a simple multicellular life cycle. PhD thesis, Christian-Albrechts-Universität Kiel, Plön, Germany.
33. Gehrig SM. 2005 Adaptation of Pseudomonas fluorescens SBW25 to the air-liquid interface: a study in evolutionary genetics. PhD thesis, University of Oxford, UK.
34. Sauer K, Stoodley P, Goeres DM, Hall-Stoodley L, Burmølle M, Stewart PS, Bjarnsholt T. 2022 The biofilm life cycle: expanding the conceptual model of biofilm formation. *Nat. Rev. Microbiol.* **20**, 608–620. (doi:10.1038/s41579-022-00767-0)
35. Fukami T, Beaumont HJE, Zhang XX, Rainey PB. 2007 Immigration history controls diversification in experimental adaptive radiation. *Nature* **446**, 436–439. (doi:10.1038/nature05629)
36. McDonald MJ, Gehrig SM, Meintjes PL, Zhang XX, Rainey PB. 2009 Adaptive divergence in experimental populations of *Pseudomonas fluorescens*. IV. Genetic constraints guide evolutionary trajectories in a parallel adaptive radiation. *Genetics* **183**, 1041–1053. (doi:10.1534/genetics.109.107110)
37. Theodosiou L, Farr AD, Rainey PB. 2023 Barcoding populations of *Pseudomonas fluorescens* SBW25. *J. Mol. Evol.* **91**, 254–262. (doi:10.1007/s00239-023-10103-6)
38. Travisano M, Rainey PB. 2000 Studies of adaptive radiation using model microbial systems. *Am. Nat.* **156**, S35–S44. (doi:10.1086/303414)
39. Spiers AJ, Kahn SG, Bohannon J, Travisano M, Rainey PB. 2002 Adaptive divergence in experimental populations of Pseudomonas fluorescens. I. Genetic and phenotypic bases of wrinkly spreader fitness. *Genetics* **161**, 33–46. (doi:10.1093/genetics/161.1.33)
40. Bittleston LS, Gralka M, Leventhal GE, Mizrahi I, Cordero OX. 2020 Context-dependent dynamics lead to the assembly of functionally distinct microbial communities. *Nat. Commun.* **11**, 1440. (doi:10.1038/s41467-020-15169-0)
41. Boyer S, Hérissant L, Sherlock G. 2021 Adaptation is influenced by the complexity of environmental change during evolution in a dynamic environment. *PLoS Genet.* **17**, e1009314. (doi:10.1371/journal.pgen.1009314)
42. Karve S, Wagner A. 2022 Environmental complexity is more important than mutation in driving the evolution of latent novel traits in *E. coli*. *Nat. Commun.* **13**, 5904. (doi:10.1038/s41467-022-33634-w)
43. Hamilton WD, May RM. 1977 Dispersal in stable habitats. *Nature* **269**, 578–581. (doi:10.1038/269578a0)
44. Rosenzweig RF, Sharp RR, Treves DS, Adams J. 1994 Microbial evolution in a simple unstructured environment: genetic differentiation in *Escherichia coli*. *Genetics* **137**, 903–917. (doi:10.1093/genetics/137.4.903)
45. Treves DS, Manning S, Adams J. 1998 Repeated evolution of an acetate-crossfeeding polymorphism in long-term populations of *Escherichia coli*. *Mol. Biol. Evol.* **15**, 789–797. (doi:10.1093/oxfordjournals.molbev.a025984)
46. Rainey PB, Kerr B. 2010 Cheats as first propagules: a new hypothesis for the evolution of individuality during the transition from single cells to multicellularity. *Bioessays* **32**, 872–880. (doi:10.1002/bies.201000039)
47. Popat R, Crusz SA, Messina M, Williams P, West SA, Diggle SP. 2012 Quorum-sensing and cheating in bacterial biofilms. *Proc. R. Soc. B* **279**, 4765–4771. (doi:10.1098/rspb.2012.1976)
48. Boyle KE, Heilmann S, van Ditmarsch D, Xavier JB. 2013 Exploiting social evolution in biofilms. *Curr. Opin. Microbiol.* **16**, 207–212. (doi:10.1016/j.mib.2013.01.003)
49. Lind PA, Farr AD, Rainey PB. 2015 Experimental evolution reveals hidden diversity in evolutionary pathways. *eLife* **4**, e07074. (doi:10.7554/eLife.07074)
50. Lind PA, Farr AD, Rainey PB. 2017 Evolutionary convergence in experimental Pseudomonas populations. *ISME J.* **11**, 589–600. (doi:10.1038/ismej.2016.157)
51. Nichols BS, Leubner-Metzger G, Jansen VAA. 2020 Between a rock and a hard place: adaptive sensing and site-specific dispersal. *Ecol. Lett.* **23**, 1370–1379. (doi:10.1111/ele.13564)
52. Duckworth RA, Belloni V, Anderson SR. 2015 Cycles of species replacement emerge from locally induced maternal effects on offspring behavior in a passerine bird. *Science* **347**, 875–877. (doi:10.1126/science.1260154)
53. Harrison RG. 1980 Dispersal polymorphisms in insects. *Annu. Rev. Ecol. Syst.* **11**, 95–118. (doi:10.1146/annurev.es.11.110180.000523)
54. King EO, Ward MK, Raney DE. 1954 Two simple media for the demonstration of pyocyanin and fluorescin. *J. Lab. Clin. Med.* **44**, 301–307.

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026

55. Bantinaki E, Kassen R, Knight CG, Robinson Z, Spiers AJ, Rainey PB. 2007 Adaptive divergence in experimental populations of Pseudomonas fluorescens. III. Mutational origins of wrinkly spreader diversity. *Genetics* **176**, 441–453. (doi:10.1534/genetics.106.069906)
56. Thielicke W, Sonntag R. 2021 Particle image velocimetry for MATLAB: accuracy and enhanced algorithms in PIVlab. *J. Open Res. Softw.* **9**, 12. (doi:10.5334/jors.334)
57. Eden M. 1961 A two-dimensional growth process (ed. J Neyman). In *Proceedings of the Fourth Berkeley Symposium on Mathematical Statistics and Probability*, vol. 4, pp. 223–239, Berkeley, CA: University of California Press.
58. Gibson MA, Bruck J. 2000 Efficient exact stochastic simulation of chemical systems with many species and many channels. *J. Phys. Chem. A* **104**, 1876–1889. (doi:10.1021/jp993732q)
59. Anderson DF. 2007 A modified next reaction method for simulating chemical systems with time dependent propensities and delays. *J. Chem. Phys.* **127**. (doi:10.1063/1.2799998)
60. Karita Y. 2025 Context-dependent adaptation in structured environments [Data set]. Zenodo. (doi:10.5281/zenodo.15735347)
61. Karita Y. 2025 Context-dependent adaptation in structured environments [Data set]. BioImage Archive. (doi:10.6019/S-BIAD2117)
62. Karita Y, Rodríguez-Sánchez GT, Brambilla E, Hernandez-Beltran JCR, Schwarz M, Rainey PB. 2026 Supplementary material from: Context-Dependent Adaptation in Structured Environments. Figshare. (doi:10.6084/m9.figshare.c.8229719)

Downloaded from http://royalsocietypublishing.org/rspb/article-pdf/doi/10.1098/rspb.2025.2004/5683302/rspb.2025.2004.pdf
by Max Planck Institute for Evolutionary Biology user
on 18 March 2026